# Practice workbook · IPv4 and IPv6: sizes, formats and why IPv6 exists

Why IPv6 was introduced, bit lengths of IPv4 and IPv6, and relating the hexadecimal group structure of an IPv6 address to its 128-bit size.

3 question parts: 1 worked-example retry and 2 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: CSC-T56-12

# 1. Question 4 · part e · 3 marks

2024 JUN · Paper 01

Worked-example retry · Attempt part e only.

**4** Peter and Marie are employees at a research facility.

(a) Peter is creating a predator-prey population simulation of rabbits and foxes.

He creates rules to simulate the movement and feeding patterns for each animal over time.

One benefit of the predator-prey simulation is that it is simpler than monitoring real life animal populations.

Explain **one other** benefit of using this simulation.

**(2)**

(b) Identify the computational model in which several computer systems work together to achieve a goal.

**(1)**

- ☒ **A** Sequential model
- ☒ **B** Parallel model
- ☒ **C** Multi-agent model
- ☒ **D** Input-process-output model

(c) Employees at the research facility use a biometric identifier to login to the network.

Describe **one other** security measure that could be used to ensure that an employee is who they claim to be.

**(2)**

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# 1. Question 4 · part e · 3 marks

2024 JUN · Paper 01

Worked-example retry · Attempt part e only.

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(d) Marie is a researcher in nanotechnology.

(i) State what is meant by the term **nanotechnology**.

(1)

(ii) Give **one** application of nanotechnology.

(1)

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(e) Marie has a mobile phone.

The address of the device is:

2024:0eb8:3c4d:0015:0000:0000:1a2f:1a2b

Describe the relationship between the structure of this address and the total number of bits required to store it.

(3)

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# 1. Question 4 · part e · 3 marks

2024 JUN · Paper 01

Worked-example retry · Attempt part e only.

(f) The research facility is considering introducing cameras and facial recognition software to monitor the work and movement of people on and around its site.

Discuss the privacy issues that need to be considered when deciding whether or not to install this new system.

**(6)**

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**(Total for Question 4 = 16 marks)**

## 1. Question 4 · part e · 3 marks · Working and answer

Working and answer

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## 2. Question 1 · part d · 2 marks

2020 NOV · Paper 01

Further practice · Attempt part d only.

**1** Upeka manages the network in her office building.

(a) Upeka uses the network to install software updates on all the office computers.

Give **two other** reasons why computers are connected in a network.

**(2)**

1 .....

.....

2 .....

.....

(b) The network uses wired and wireless connectivity.

(i) State **two** advantages of using wired rather than wireless connectivity.

**(2)**

1 .....

.....

2 .....

.....

(ii) State **two** disadvantages of using wired rather than wireless connectivity.

**(2)**

1 .....

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2 .....

.....

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2

P 6 1 8 8 4 R R A 0 2 2 0

## 2. Question 1 · part d · 2 marks

2020 NOV · Paper 01

Further practice · Attempt part d only.

(c) Some people confuse the terms 'internet' and 'world wide web'.

Describe the difference between the internet and the world wide web.

**(2)**

(d) Networks use both IPv4 and IPv6 addresses to identify connected devices.

Explain why IPv6 addressing was introduced.

**(2)**

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## 2. Question 1 · part d · 2 marks

2020 NOV · Paper 01

Further practice · Attempt part d only.

(e) Network layouts can be based upon a number of different network topologies.

**Figure 1** shows a network topology.

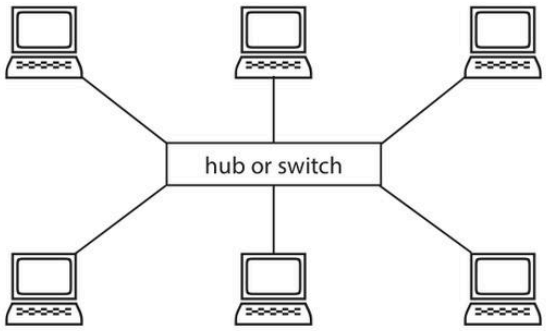



**Figure 1**

(i) State the name of this network topology.

(1)

(ii) State **one** benefit of using this network topology.

(1)

(iii) State **one** drawback of using this network topology.

(1)

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## 2. Question 1 · part d · 2 marks · Working and answer

Working and answer

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## 3. Question 4 · part b · 2 marks

2023 JUN · Paper 01

Further practice · Attempt part b only.

**4** Data packets travel across networks from one device to another.

(a) Identify the measurement of network speeds.

(1)

- ☒ **A** Mebibits per second
- ☒ **B** Mebibytes per second
- ☒ **C** Megabits per second
- ☒ **D** Megabytes per second

(b) Data packets contain the addresses of the sender and the receiver.

Complete the table to give the number of bits that make up each type of network address.

(2)

| Type | Example | Number of bits |
|---|---|---|
| IPv4 | 192.169.0.3 | |
| IPv6 | 1050:a500:00c0:0440:0006:0300:700d:436f | |

(c) Data packets travel over physical media.

(i) Describe **one** difference between the media used by a wired network and a wireless network.

(2)

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## 3. Question 4 · part b · 2 marks · Working and answer

Working and answer

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